Reg. No.															

B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, DECEMBER 2025

Sixth Semester

21ECC302T - ANALOG AND DIGITAL COMMUNICATION

(For the candidates admitted during the academic year 2021 - 2022 to 2024 - 2025)

Note:

i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40^{th} minute.

ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 hours **Max. Marks: 75**

PART - A (20 × 1 = 20 Marks)

Answer **all** Questions

	Question	Marks	CO	BL
1.	For Amplitude Modulation, Emitter modulator ________ (A) Operates in class C mode (B) Has a low efficiency (C) Output power is high (D) Operates in class B mode	1	1	1
2.	If modulation index of an AM wave is increased from 1.5 to 2, then the transitted power ________ (A) Increases by 50% (B) Remains same (C) Increases by 20% (D) Increases by 41%	1	1	2
3.	Envelope Detector is a/an ________ (A) Asynchronous Detector (B) Synchronous Detector (C) Product Demodulator (D) Coherent detector	1	1	1
4.	What is the bandwidth of a FM wave when maximum allowed deviation is 50KHz and the modulating signal has a frequency of 15KHz? (A) 260 KHz (B) 50 KHz (C) 130 KHz (D) 65 KHz	1	1	3
5.	What is the purpose of the Automatic Volume Control (AVC) in a receiver? (A) To amplify the audio signal to a desired level (B) To modulate the signal at the receiver (C) To filter out unwanted frequency signals (D) To automatically adjust the volume to compensate for noise and variations in the signal	1	2	1
6.	The ability of a radio receiver to select a desired signal and reject unwanted signals on other frequencies is known as (A) Selectivity (B) Fidelity (C) Figure of merit (D) Sensitivity	1	2	2
7.	A superheterodyne receiver receives signal within frequency range of 120 to 180 MHz, then the required Intermediate frequency is ________ (A) 60 MHz (B) 20 MHz (C) 30 MHz (D) 45 MHz	1	2	3
8.	What is the main limitation of a standard TRF receiver? (A) Requires variable Q-factor (B) Variable selectivity and fidelity across the frequency range (C) Overly complex design with many components (D) Inability to tune to different frequencies	1	2	2

9. What type of digital modulation is widely used for digital data transmission? 1 3 1

(A) Pulse position modulation
(B) Pulse width modulation
(C) Pulse amplitude modulation
(D) Pulse code modulation

10. Which of the following is false with respect to pulse position modulation? 1 3 2

(A) Can be transmitted in narrowband
(B) Pulse width changes in accordance with the amplitude of modulating signal
(C) Pulse is narrow
(D) Modulates a high frequency carrier

11. What is the output voltage of a μ-law compander with $\mu = 255$ and a maximum input range of 1 V, when the input voltage is 0.25 V? 1 3 3

(A) 0.75 V
(B) 0.25 V
(C) 0.4 V
(D) 0.5 V

12. The use of non uniform quantization leads to 1 3 2

(A) Increase in SNR for low level signals
(B) Reduction in transmission bandwidth
(C) Increase in maximum SNR
(D) Simplification of quantization process

13. In an M-ary PSK system, what happens to the probability of error as the value of M (the number of phases) increases, assuming constant E_b / N_o 1 4 1

(A) The probability of error decreases.
(B) The probability of error increases.
(C) The probability of error is zero
(D) The probability of error remains the same.

14. The probability of error for coherent BPSK is given by which expression? 1 4 1

(A) $$P_e = \frac{1}{2} \exp\left(-\frac{E_b}{N_0}\right)$$
(B) $$P_e = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$$
(C) $$P_e = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$
(D) $$P_e = Q\left(\sqrt{\frac{E_b}{2N_0}}\right)$$

15. When comparing coherent BFSK and coherent BPSK in an AWGN channel, which statement is true? 1 4 2

(A) Their performance is not comparable.
(B) They have the same BER performance.
(C) Coherent BFSK has a better BER performance than coherent BPSK.
(D) Coherent BPSK has a better BER performance than coherent BFSK.

16. In M-ary FSK, what happens to the probability of error as M increases, assuming orthogonal signaling? 1 4 2

(A) It decreases.
(B) It increases.
(C) It remains constant.
(D) It depends on the channel.

17. In the Shannon-Hartley theorem, $C = B \log_2(1 + S / N)$, what happens to the channel capacity (C) as the bandwidth (B) approaches infinity? 1 5 2

(A) C approaches infinity.
(B) C approaches zero
(C) C remains constant
(D) C approaches to finite value

18. A channel has a bandwidth of 1 MHz and an SNR of 1023. What is the channel capacity (C) ? (A) 10 Mbps (B) 100 Mbps (C) 1 Mbps (D) 10 Gbps		1	5	3
19. The mutual information, I(X;Y), is defined as: (A) I(X;Y) = H(X,Y) (B) I(X;Y) = H(X) - H(X|Y) = H(Y) - H(Y|X) (C) I(X;Y) = H(Y) - H(X|Y) (D) I(X;Y) = H(X) - H(Y|X)		1	5	2
20. Entropy is a measure of (A) The maximum data rate of a channel. (B) The signal-to-noise ratio of a channel. (C) The number of errors in a transmitted message. (D) The average amount of information per source output.		1	5	2

PART - B (5 × 8 = 40 Marks)

Answer **all** Questions

	Marks	CO	BL
21. a) With neat diagram, explain the operation of Phase Discriminator Method to generate the Single Side Band modulation.	8	1	2
(OR)			
b) Explain the operation and general expression for Non Linear modulation with Balance Modulator to generate AM signals.	8	1	2
22. a) Explain the superheterodyne FM broadcast receiver with proper block diagram and also discuss each block of it.	8	2	2
(OR)			
b) Explain the implementation of AM transmitter using low-level and high-level modulation techniques with neat block diagrams.	8	2	2
23. a) Derive a general expression for evaluating the Probability of Error when a signal is absent as well as present, on choosing the threshold level "a".	8	3	4
(OR)			
b) (i) Prove that the Quantization noise power for PCM system is S2 / 12 where S is a step size (4 Marks) (ii) Describe briefly about the generation of PAM, PWM and PPM (4 Marks)	8	3	4
24. a) Discuss the generation and signal space diagram of QPSK.	8	4	1
(OR)			
b) Briefly explain about the generation and detection of QAM signals with relevant block diagram.	8	4	1
25. a) Derive the expression of Channel Capacity of Continious Noisy Channel using Shannon Hartley Theorem, with infinte bandwidth	8	5	4
(OR)			
b) Find the bandwidth of a picture signal in TV when following data are specified: (i) TV picture consists of 3 X 10 5 small picture elements. (ii) There are 10 different brightness levels which are equi-likely to occur. (iii) Number of frames per second is 30. (iv) SNR = 30 dB	8	5	4

PART - C ($1 \times 15 = 15$ Marks)

Answer **any 1** Question

		Marks	CO	BL
26.	(i) Give the drawback of PSK demodulation. (2 Marks) (ii) Describe with neat diagram, how DPSK avoids the use of Synchronous demodulation in PSK (8 marks) (iii) Describe the modulation and demodulation processes of Frequency Shift Keying (FSK) with neat diagrams and appropriate mathematical expressions.(5 marks)	15	4	3
27.	Design a binary Huffman's coding for a discrete source of 3 independent symbols - α, β and γ with probabilities of 0.9, 0.08, 0.02 respectively. It is interested to design a binary first order extension code i.e 2 symbols taken at a time for the same discrete source. Calculate (i) Coding Efficiency (4 marks) (ii) Compression Ratio (3 marks) (iii) Average length of the code (4 marks) (iv) Entropy of the source (4 Marks)	15	5	4

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